

Sales Analytics Dashboard using SQL Server, MySQL & Power BI

Project Overview

This project demonstrates the development of an end-to-end Business Intelligence reporting solution using SQL Server, MySQL, and Power BI.

The project focuses on importing, transforming, validating, and visualizing business data to generate actionable insights and KPI-based dashboards for business optimization.

Project Objective

The primary objective of this project is to:

- Build interactive Power BI dashboards for business analysis
 - Perform data cleaning and transformation using SQL and Power Query
 - Create KPI measures and performance indicators using DAX
 - Transition reporting workflow from Test Environment to Production Environment
 - Integrate SQL Server and MySQL databases with Power BI
 - Publish business reports to Power BI Workspace
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Tools & Technologies Used

- SQL Server
 - MySQL Workbench
 - Power BI Desktop
 - SQL
 - DAX
 - Power Query
 - Power BI Service
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Project Workflow

Step 1 — Environment Setup

- Installed and configured SQL Server and MySQL Workbench
 - Prepared test environment for data processing and reporting
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Step 2 — Data Import and Preparation

- Imported datasets into SQL Server

- Organized relational data tables
 - Performed data cleaning and preprocessing
 - Prepared transformed datasets for reporting
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Step 3 — SQL Data Transformation

- Applied joins and filtering operations
 - Structured business data for analysis
 - Validated transformed datasets before reporting
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Step 4 — Power BI Integration

- Connected Power BI Desktop with SQL Server
 - Imported transformed datasets into Power BI
 - Configured relationships and data models
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Step 5 — KPI and Dashboard Development

- Created DAX measures and KPI indicators
 - Developed interactive dashboard pages
 - Added filters, KPI cards, and business visuals
 - Designed dashboards for operational and financial analysis
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Step 6 — Transition from Test to Production

- Transitioned reporting workflow from Test Environment to Production Environment
 - Updated database connections and transformed reports
 - Validated production data accuracy
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Step 7 — MySQL Integration

- Installed MySQL Connector
 - Imported production data into MySQL database
 - Connected Power BI with MySQL data source
 - Transitioned reports using Power Query Advanced Editor
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Step 8 — Report Publishing

- Published dashboards to Power BI Workspace
 - Verified reporting functionality and accessibility
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Dashboard Metrics

Page 1 — Demand & Availability Analysis

The first dashboard page focuses on operational demand and supply analysis using KPI indicators such as:

- Average Demand per Day
- Average Availability per Day
- Total Supply Shortage

This dashboard helps identify supply-demand gaps and operational shortages to support business optimization.

Page 2 — Profit & Loss Analysis

The second dashboard page focuses on financial performance metrics including:

- Total Profit
- Total Loss
- Average Daily Loss

This dashboard helps analyze operational profitability and business losses.

Data Transformation & Validation

Data transformation and validation were performed using Power Query Editor in Power BI:

- Data quality validation
- Column profiling
- Null value handling
- Data filtering and transformation
- Data consistency checks

The transformed data was then used for dashboard reporting and KPI generation.

Dashboard Design Features

The dashboard includes:

- KPI Cards
 - Interactive Filters
 - Multi-Page Dashboard Navigation
 - Professional Dark-Themed UI
 - Business Optimization Reporting
 - Real-Time Metrics Visualization
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Key Features

- End-to-End BI Reporting Workflow
 - SQL-Based Data Processing
 - Interactive Power BI Dashboards
 - KPI and Performance Tracking
 - DAX Measure Creation
 - Test to Production Environment Transition
 - MySQL Database Integration
 - Data Validation and Reporting
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Business Insights Generated

The dashboard provides insights into:

- Demand vs Availability Analysis
 - Supply Shortage Monitoring
 - Profit and Loss Tracking
 - Operational Performance
 - Business KPI Monitoring
 - Data-Driven Decision Support
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Challenges Faced

During the project, the following challenges were addressed:

- Managing data inconsistencies
- Preparing relational datasets for reporting
- Transitioning reports between environments

- Validating transformed data accuracy
 - Configuring database integrations
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Outcome

Successfully developed an end-to-end Business Intelligence reporting solution integrating SQL Server, MySQL, and Power BI for reporting, KPI monitoring, and business analysis.

The project demonstrates practical knowledge in:

- SQL Querying
 - Data Transformation
 - Power BI Reporting
 - DAX Calculations
 - Database Integration
 - BI Workflow Management
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Additional Resources

All dashboard screenshots, Power Query screenshots, and project visuals are available in the screenshots folder of this repository.

Author

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